Analysis 2. Test 2

① a) $(x>0)$ $\int\left(\frac{1}{x^2}-\sqrt{1+\frac{1}{x}}\right)^2 dx = \int\left(\frac{1}{x^4}-\frac{2}{x^2}\cdot\sqrt{1+\frac{1}{x}}+1+\frac{1}{x}\right)dx =$

$= \int\frac{1}{x^4}dx + \int 1\,dx + \int\frac{1}{x}dx + 2\int\underbrace{\left(1+\frac{1}{x}\right)'}_{f'}\cdot\underbrace{\left(1+\frac{1}{x}\right)^{\frac{1}{2}}}_{f^\alpha}dx =$

$\int f'\cdot f^\alpha$

$= \frac{x^{-3}}{-3} + x + \ln(x) + 2\cdot\frac{\left(1+\frac{1}{x}\right)^{\frac{3}{2}}}{\frac{3}{2}} + C =$

$= \boxed{-\frac{1}{3x^3} + x + \ln x + \frac{4}{3}\cdot\sqrt{\left(1+\frac{1}{x}\right)^3} + C \quad (C\in\mathbb{R})}$

b) $\int(1-2x)\cdot\operatorname{arctan} x\,dx \quad (x\in\mathbb{R}) =$

int. par.

$= \int\underbrace{(x-x^2)'}_{g'}\cdot\underbrace{\operatorname{arctan} x}_{f}\,dx = (x-x^2)\cdot\operatorname{arctan} x -$

$-\int(x-x^2)\frac{1}{1+x^2}dx = (x-x^2)\operatorname{arctan} x +$

$+\int\frac{x^2-x}{x^2+1}dx = (x-x^2)\operatorname{arctan} x + \int\left(\frac{x^2+1}{x^2+1}-\frac{x}{x^2+1}-\frac{1}{x^2+1}\right)dx$

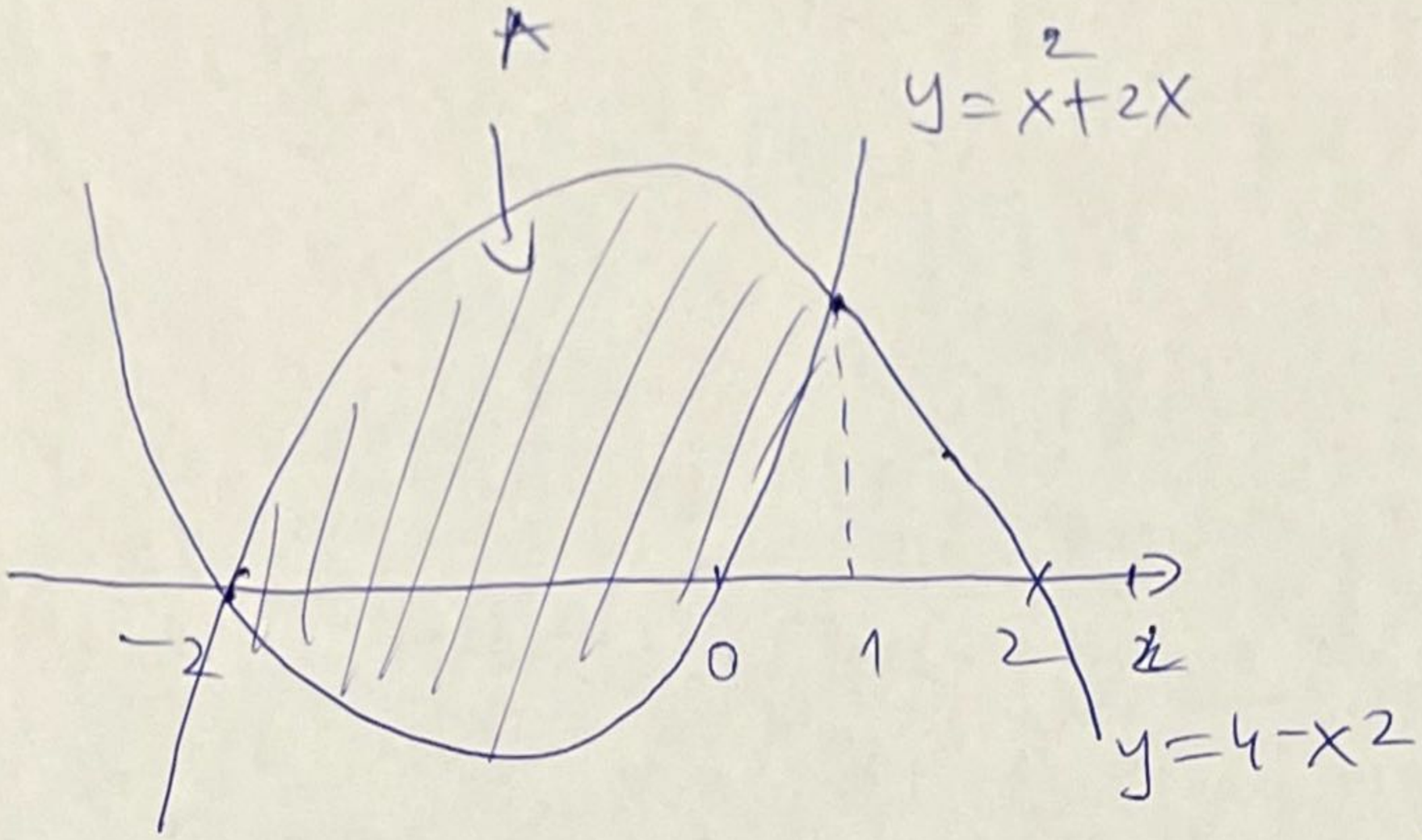
$= (x-x^2)\operatorname{arctan} x + \int 1\,dx - \frac{1}{2}\cdot\int\frac{(x^2+1)'}{x^2+1}dx - \int\frac{1}{x^2+1}dx$

$= \underbrace{(x-x^2)\operatorname{arctan} x}_{u\cdot v} + x - \frac{1}{2}\underbrace{\ln(x^2+1)}_{\circledast} - \operatorname{arctan} x + C =$

$= \boxed{(x-x^2-1)\operatorname{arctan} x + x - \frac{1}{2}\ln(x^2+1) + C}$

$(C\in\mathbb{R})$

(2)



Int. points: $x^2 + 2x = 4 - x^2 \iff 2x^2 + 2x - 4 = 0$

$\iff x^2 + x - 2 = 0 \iff (x+2)(x-1) = 0 \iff$

$\boxed{x_1 = -2}$, $\boxed{x_2 = 1}$.

$$A = \int_{-2}^{1} \left(4 - x^2 - (x^2 + 2x)\right) dx = \int_{-2}^{1} (-2x^2 - 2x + 4)\, dx =$$

$$= \left[-\frac{2x^3}{3} - x^2 + 4x\right]_{-2}^{1} = -\frac{2}{3} - 1 + 4 - \left(\frac{16}{3} - 4 - 8\right)$$

$$= \frac{-2}{3} - \frac{16}{3} + 3 + 12 = 15 - 6 = \boxed{9}$$

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(3) $f(x) := \frac{1}{x} \cdot \sqrt{\frac{x+1}{x-1}}$ $\left(x \in [2,3]\right)$, $f \in C[2,3] \Rightarrow$

Volume $= \pi \cdot \int_2^3 f^2(x)\, dx = \pi \int_2^3 \frac{1}{x^2} \cdot \frac{x+1}{x-1}\, dx$

$$\frac{x+1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} \qquad (A,B,C \in \mathbb{R} = ?)$$

$$x+1 = Ax(x-1) + B(x-1) + Cx^2 \iff$$

$$x+1 = (A+C)x^2 + (-A+B)\cdot x - B \iff$$

$$\begin{cases} A+C=0 \\ A+B=1 \\ -B=1 \end{cases} \Rightarrow \boxed{B=-1}\ ;\ \boxed{A=-2} \Rightarrow \boxed{C=2}$$

So:

$$V = \pi \cdot \int_2^3 \left(-\frac{2}{x} - \frac{1}{x^2} + \frac{2}{x-1}\right) dx =$$

$$= \pi \left[\underset{\text{on } [2,3]}{\underset{\oplus}{-2\cdot \ln x}} + \frac{1}{x} + \underset{\text{on } [2,3]}{\underset{\oplus}{2\ln(x-1)}} \right]_2^3 =$$

$$= \pi \cdot \left(-2\ln 3 + \frac{1}{3} + \underline{2\ln 2} - \left(\underline{-2\ln 2} + \frac{1}{2} + \underset{0}{2\ln 1} \right) \right)$$

$$= \pi \left(4\ln 2 - 2\ln 3 + \frac{1}{3} - \frac{1}{2} \right) =$$

$$= \boxed{\pi \left(\ln \frac{16}{9} - \frac{1}{6} \right)};$$

(4) $\boxed{f(x,y) = \frac{x^3}{3} - 3x^2 - 4xy + y^2 \quad ((x,y) \in \mathbb{R}^2)}$

Local extrema = ?

Sol.

$$\begin{cases} \partial_1 f(x;y) = x^2 - 6x - 4y = 0 \\ \partial_2 f(x;y) = -4x + 2y = 0 \Rightarrow y = 2x \end{cases} \Rightarrow$$

$$x^2 - 6x - 4\cdot(2x) = 0$$

$$x^2 - 6x - 8x = 0$$

$$x^2 - 14x = 0$$

$$x(x-14) = 0 \Leftrightarrow \begin{cases} x_1 = 0 \\ y_1 = 0 \end{cases} \text{ or } \begin{cases} x_2 = 14 \\ y_2 = 28 \end{cases}$$

Stationary points: $\boxed{P_1(0;0);\ P_2(14;28)}$

And:

$$f''(x;y) = \begin{pmatrix} 2x-6 & -4 \\ -4 & 2 \end{pmatrix} \quad ((x;y) \in \mathbb{R}^2) \Rightarrow$$

$$f''(0;0) = \begin{pmatrix} -6 & -4 \\ -4 & 2 \end{pmatrix} : \Delta_1 = -6 < 0$$

$$\Delta_2 = -12 - 16 = -28 < 0$$

$\Rightarrow$ $\boxed{(0;0) \text{ is not local extr. point}}$, $\boxed{\text{it's a saddle p.}}$

$$f''(14;28) = \begin{pmatrix} -2\cdot 14 - 6 & -4 \\ -4 & 2 \end{pmatrix} = \begin{pmatrix} -34 & -4 \\ -4 & 2 \end{pmatrix}$$

So: $\Delta_1 = -34 < 0$

$$\Delta_2 = -68 - 16 = -84 < 0 \Rightarrow$$

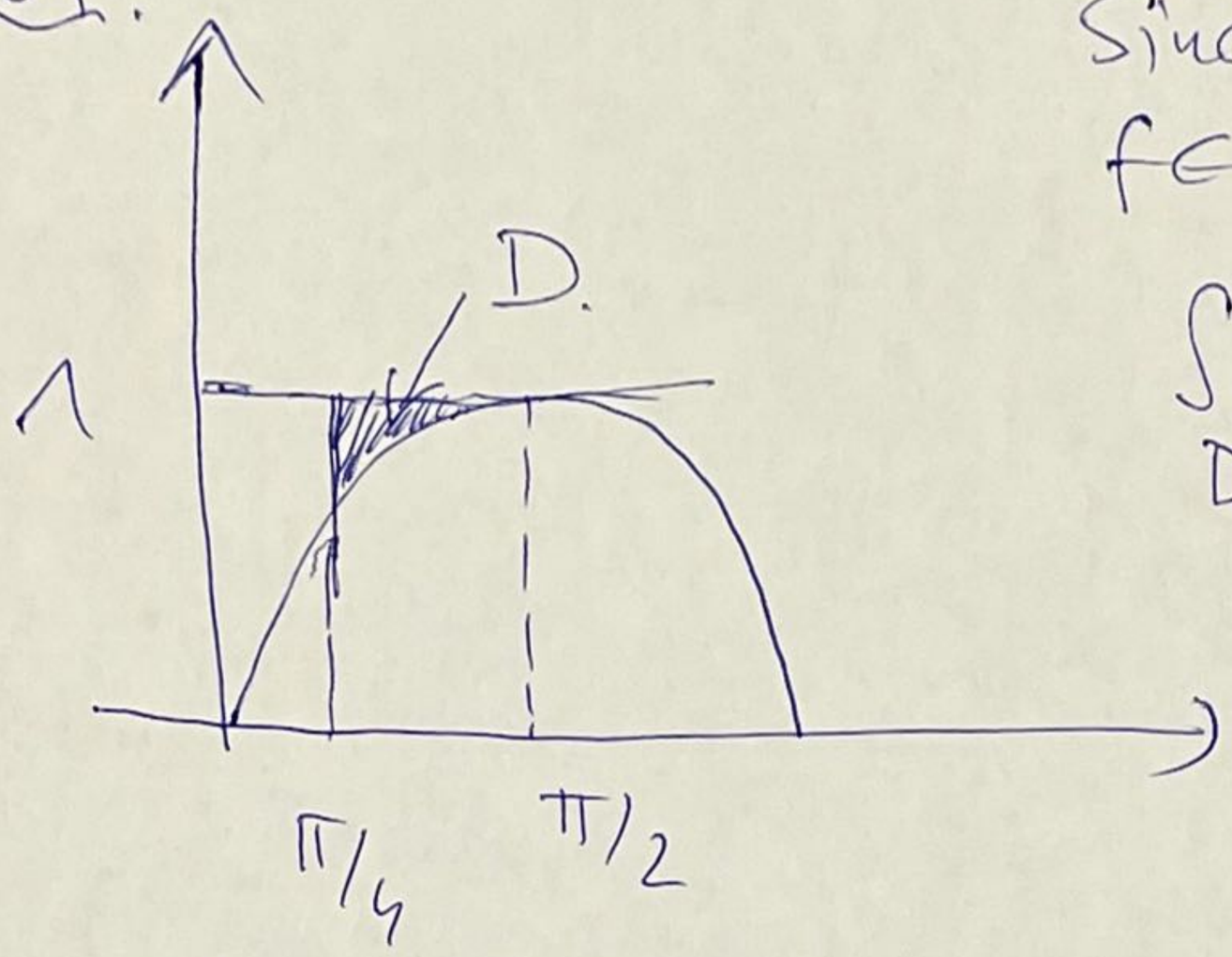
$\boxed{(14;28) \text{ is not local extr. point}}$.

5. $f(x,y) = \frac{\sin^2 x}{y^2}$ $\left((x,y) \in \mathbb{R} \times (0;+\infty)\right)$.

$$D := \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; \frac{\pi}{4} \le x \le \frac{\pi}{2},\ \sin x \le y \le 1 \right\}$$

$$\iint_D f = ?$$

Sol:



Since $f \in C(D) \Rightarrow$
$f \in R(D)$ and

$$\iint_D f = \int_{\pi/4}^{\pi/2} \int_{\sin x}^{1} \frac{\sin^2 x}{y^2}\, dy\, dx$$

$$= \int_{\pi/4}^{\pi/2} \sin^2 x \left(\int_{\sin x}^{1} \frac{1}{y^2}\, dy \right) dx = \int_{\pi/4}^{\pi/2} \sin^2 x \cdot \left[-\frac{1}{y} \right]_{\sin x}^{1} dx =$$

$$= \int_{\pi/4}^{\pi/2} \sin^2 x \cdot \left(\frac{1}{\sin x} - 1 \right) dx = \int_{\pi/4}^{\pi/2} \left(\sin x - \sin^2 x \right) dx =$$

$$= \int_{\pi/4}^{\pi/2} \left(\sin x - \frac{1 - \cos 2x}{2} \right) dx =$$

$$= \left[-\cos x - \frac{x}{2} + \frac{\sin 2x}{4} \right]_{\pi/4}^{\pi/2} =$$

$$= -\overbrace{\cos\frac{\pi}{2}}^{0} - \frac{\pi}{4} + \overbrace{\frac{\sin\pi}{4}}^{0} - \left(-\cos\frac{\pi}{4} - \frac{\pi}{8} + \frac{\overbrace{\sin\frac{\pi}{2}}^{1}}{4} \right) =$$

$$= -\frac{\pi}{4} + \frac{\sqrt{2}}{2} + \frac{\pi}{8} - \frac{1}{4} = \boxed{\frac{2\sqrt{2}-1}{4} - \frac{\pi}{8}}.$$